

Multiple Choice

1. **Answer:** D

Options: A) Decreased calcium excretion in the urine; B) Increased risk of fracture; C) Decreased dietary calcium absorption; D) Negative calcium balance and bone mineral loss

Note: Correct option: D - Negative calcium balance and bone mineral loss

2. **Answer:** D

Options: A) O₂ consumption alone; B) CO₂ production alone; C) Breath collection; D) Total urinary nitrogen excretion alone

Note: Correct option: D - Total urinary nitrogen excretion alone

3. **Answer:** A

Options: A) <10% of total daily energy; B) <12%; C) <15%; D) <16%

Note: Correct option: A - <10% of total daily energy

4. **Answer:** C

Options: A) Normal energy intake, normal body weight and hyperthyroidism; B) Obesity, excess energy intake, normal growth and hypoinsulinaemia; C) Obesity, abnormal growth, hypothyroidism, hyperinsulinaemia; D) Underweight, abnormal growth, hypothyroidism, hyperinsulinaemia

Note: Correct option: C - Obesity, abnormal growth, hypothyroidism, hyperinsulinaemia

5. **Answer:** A

Options: A) DNMT 1; B) DNMT 3a; C) DNMT 3b; D) DNMT 3L

Note: Correct option: A - DNMT 1

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Starch mainly contains amylose, which is a polymer of glucose with alpha 1,4-linkages'.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
10. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** Increased insulin secretion in the fed state.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.
12. **Answer:** High dose β -carotene supplements. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key